Structured data sources for the Fessi waste disposal knowledge graph

The most actionable finding is PBSeasy (PBS Network GmbH), the industry-standard product data platform for European office/stationery products with 1 million+ items from 150+ manufacturers — a source almost invisible to general web searches but central to the German PBS (Papier, Büro, Schreibwaren) sector. Combined with the Blauer Engel Fo.KUS database (8,850+ EAN-coded eco-certified products, available via XML interface for free), Forum CSR My Sitetitle the EU Ecolabel open data CSV (109,096 products with EAN13 codes), data.europa.eu ECHA's SCIP database (15 million+ substance-of-concern factsheets), Chementors and Icecat's open product catalog (26 million+ datasheets with toner/cartridge compatibility), these five sources form a viable foundation for populating Fessi's Graph-RAG knowledge graph. Material composition remains the critical gap — no single structured database provides "what office products are made of" — but a pipeline combining SCIP data, manufacturer safety data sheets, and eCl@ss properties can partially close it.

Executive summary: the five highest-value sources

The research investigated **40+ potential sources** across BMEcat catalogs, toner databases, eco-labels, classification standards, procurement portals, and material composition databases. Five stand out as immediately actionable:

Rank	Source	Why it matters	Format	Access	Records
1	**PBSeasy (PBS Network)**	Only industry-wide structured product data platform for office/stationery in Europe	REST API, JSON, Excel	Paid B2B	**1M+ products**
2	**Blauer Engel Fo.KUS DB**	EAN/GTIN-coded German eco-label data with XML interface, daily updates	XML, file download	**Free** (request)	**8,850+ EANs**
3	**EU Ecolabel ECAT**	Open data CSV with EAN13 barcodes, licence info, company data	CSV, API	**Free** open data	**109,096 products**
4	**ECHA SCIP Database**	Only source linking specific products to hazardous substance composition	Web portal (IUCLID)	**Free**	**15M+ factsheets**
5	**Icecat Open Catalog**	Largest open product data source with toner compatibility, EAN/GTIN, specs	JSON/XML/CSV API	Free tier + paid	**26M+ datasheets**

PBSeasy is the hidden backbone of German office supply data

PBS Network GmbH (Stuttgart) operates the industry-standard data exchange platform for the entire European

PBS sector. (nexoma) This source is virtually invisible to generic web searches but represents the single most comprehensive structured product data source for office supplies.

What PBSeasy contains: Complete product master data (descriptions, images, marketing copy) for **1 million+ products** from **150+ European manufacturers** including major brands like LEITZ ACCO Brands and Schneider Novus. (Retail Optimiser) The platform delivers real-time net purchase prices, live stock/inventory levels, and **eCl@ss classification** (Segment 24: "Office product, facility and technic, papeterie"). (Manutan) Data undergoes **200+ quality checks** before delivery. Access methods include REST API, custom PBS connectors, Excel upload/download, and ERP/PIM/web shop integration. The platform connects to GDSN via a b-synced partnership with Bayard Consulting.

Limitations: PBSeasy is a commercial B2B platform — it requires dealer/retailer registration and is paid. Material composition is not a primary data field, though eCl@ss properties may include material attributes for certain product classes. Negotiating research/academic access is essential.

Relevance: 5/5. Provider: PBS Network GmbH. URL: pbsnetwork.eu. Geographic scope: EU (strongest DACH). Languages: DE, EN, FR. Update frequency: continuous. License: commercial B2B.

Blauer Engel and EU Ecolabel provide the best free structured data

Two eco-label databases offer immediate, machine-readable data with product identifiers that can serve as anchors in the knowledge graph.

Fo.KUS e.V. / Blauer Engel database is maintained by Fo.KUS e.V. on behalf of the Umweltbundesamt (UBA). It contains **8,850+ EAN/GTIN codes** of Blauer Engel-certified consumer products, available via an **XML interface** and file download with **daily updates**. Access is free — a simple informal request ("formlose Anfrage") to blauerengel@fo-kus.info suffices. (bevh +2) The database covers printers/MFDs (Forum CSR) (**1,474 laser/LED devices**, 1,218 multifunction devices, 133 inkjet devices), (blauer-engel) paper products (150+ office papers, 117 printing papers), office supplies (66 notepads, 55 folders, 42 dividers, 34 binders, 33 envelopes), writing utensils (highlighters, ballpoint pens, fountain pens), (blauer-engel) and remanufactured toner cartridges under **DE-UZ 200**. Key product group standards: DE-UZ 14a/14b (recycled paper), (Verdi) DE-UZ 171 (printers), DE-UZ 177 (writing utensils), DE-UZ 200 (remanufactured toner). The main blauer-engel.de website lists **70,000+ products** from 1,800+ companies (Blue Angel +2) but has no public API — only web browsing. An "E-Commerce data service" page hints at structured data feeds for online retailers.

EU Ecolabel ECAT provides fully open data: a **CSV download** at data.europa.eu containing 109,096 products from 3,384 licences. Fields include product name, licence number, group code/name, **EAN13 barcode** (EXTID_TYPE), (European Commission) company name and country, VAT number, criteria decision reference, and expiration date. (data.europa.eu) An API and Power BI dashboard are also available. (European Commission) Office-relevant groups include graphic paper, cardboard, converted paper products, printed matter, and furniture. The CSV is immediately downloadable and parseable — no registration required.

Relevance: 5/5 (Fo.KUS), 4/5 (EU Ecolabel). Both are free. The EAN/GTIN fields in both databases enable cross-referencing with commercial product databases and barcode-based disposal lookups.

ECHA SCIP is the only source linking products to hazardous substances

The SCIP database (Substances of Concern In articles as such or in complex objects/Products) is mandated by the EU Waste Framework Directive (ECHA) and contains **15 million+ factsheets** from **33 million+ notifications** (Chementors) submitted by 6,000+ companies. (Greensofttech) It is the only publicly accessible database that connects specific commercial articles to their hazardous substance content.

Available fields: Article name, identifiers (brand, EAN, part numbers), article category (TARIC-based), **material category** where the SVHC is present, SVHC substance name with EC/CAS numbers, **concentration range (w/w%)**, safe use instructions including disposal stage, and product hierarchy for complex objects (bill-of-materials structure). (ECHA) The critical limitation is that SCIP only covers articles containing SVHCs above **0.1% w/w** (ECHA) — so standard paper, basic pens, and folders without hazardous additives will not appear. Office products most likely present include toner cartridges, solvent-based markers/highlighters, correction fluids, adhesives, PVC binders/folders (phthalate plasticizers), and electronics.

Access constraints: Free public web search portal at echa.europa.eu/scip-database, but **no bulk download API** for reading data. The submission API (S2S) uses IUCLID XML format with REST/Swagger 2.0, (ECHA) but this is for data submission only. ECHA reported "severe delays in SCIP dissemination" as of January 2026. (ECHA) Supplier identity is not disclosed in public view. (Yordas Group)

Relevance: 4/5. Format: web portal (no bulk export). Provider: European Chemicals Agency. Scope: EU-wide. Language: EN (multilingual submissions). License: public data, free access.

Toner cartridges require a multi-source pipeline — no single database exists

The highest-value product subset for waste disposal (toner/ink cartridges) has no unified database mapping **cartridge → material composition → AVV waste code**. Building this mapping requires chaining five data layers.

Layer 1 — Product identification (Icecat). Icecat's open catalog contains **26 million+ product datasheets** across 28,800+ brands, (Icecat) with extensive "Printing Supplies" coverage. Free (Open Icecat) for brand-sponsored content; (Iceclog) full access requires a paid subscription. Fields include brand, model, **EAN/GTIN/UPC barcodes**, technical specs, compatible printer models, cartridge type, color, page yield, and OEM part numbers. Available via REST API (JSON, XML, CSV) (Icecat) with max 100 queries/second. (Iceclog) No hazard or material composition data.

Layer 2 — Cross-manufacturer compatibility (TRI Resources). TRI Resources International operates the **industry gold standard** for printer supplies cross-referencing. The T3 Ink and Toner Finder API maps cartridges to compatible printers across OEM and aftermarket manufacturers (triresources) with 1,000+ supply photos and nightly automated updates. (triresources) Access is paid (setup + monthly fees) with a free 1-month trial for TRIguideOnline.

Layer 3 — Safety data sheets (OEM manufacturer portals). All major toner manufacturers maintain structured SDS repositories. **Xerox** provides an open directory listing at xerox.com/download/ehs/msds/ organized by product code with German-language PDFs — highly scrapeable. **HP** uses systematic document

IDs (e.g., c07705107). **Canon** hosts SDS at files.canon-europe.com/files/webcontent/safety-data-sheets/. Sharp, Kyocera, OKI, Brother, and Konica Minolta all publish per-product SDS. Critical finding: **most modern commercial toner cartridges are NOT classified as hazardous** under CLP/GHS — toner powder is typically styrene-acrylate polymer with encapsulated carbon black, silica, and wax, mitigating the IARC 2B classification of free carbon black.

Layer 4 — Substance-level hazard data (GESTIS + ECHA CHEM). The GESTIS database (IFA/DGUV) contains **9,400+ substances** (Wikipedia) with full GHS classification, toxicological data, (Unitar) occupational exposure limits, and WGK water hazard classes — the most authoritative German source. Free for safety purposes; commercial use and bulk transfer prohibited. (DGUV) Linking is "explicitly desired." (DGUV) ECHA CHEM provides **100,000+ REACH registration dossiers** with substance identity, physico-chemical properties, classification & labelling, and exposure scenarios — free and public. (AIHA) Key toner constituents (carbon black CAS 1333-86-4, styrene-acrylate polymers, iron oxide pigments, silica) are registered.

Layer 5 — AVV code assignment. The waste code determination (08 03 17* hazardous vs. 08 03 18 non-hazardous) depends on whether hazardous substances exceed threshold concentrations. The mapping must be constructed: identify cartridge (Icecat/EAN) → find SDS (manufacturer portals) → extract composition → cross-reference ECHA/GESTIS hazard data → apply AVV threshold rules → assign code. This pipeline is the **core value-add** Fessi would provide.

Take-back program data exists but is unstructured: HP Planet Partners publishes a "non-returnable supplies list" PDF (updated October 2025, Rev. 8) identifying cartridges with components preventing recycling — the closest thing to a recyclability database. Interzero (formerly Interseroh) operates "Der Sammeldrache" for school/business cartridge collection (Interzero) (1.4M cartridges/year) (Fraunhofer UMSICHT) but publishes no structured data. ETIRA provides market intelligence (120M toner cartridges/year in Europe, 20-25% non-OEM share) (Etira) and standards references (DIN 33870 for remanufactured B&W toner, DIN 33871 for inkjet, ISO/IEC 29142 for cartridge characterization) (Etira) but no product-level database.

Classification systems form the taxonomic bridge to waste codes

Three classification systems are available for structuring the knowledge graph's product hierarchy, with varying depth and cost.

eCl@ss (Segment 24) is the dominant standard in German industry (Wikipedia) with **~50,000 classes, 23,000 properties, and 140,000 keywords** across 31 languages. (ECLASS) (Wikipedia) Release 16.0 (November 2025) (ECLASS) added (ECLASS) (ECLASS) **material declaration aspects for all product classes**, (ECLASS) Product Carbon Footprint/Product Environmental Footprint with 24 impact categories, (ECLASS) and a first draft for **recycling information** (to be expanded in Release 17). (ECLASS) The Basic Set of Properties includes GTIN, manufacturer name, product numbers, and product type description. (ECLASS) A **free content search tool** at eclass.eu allows browsing and searching without a license. (ECLASS) Full download requires a paid license scaled by organization size. (ECLASS) eCl@ss is preparing for EU Digital Product Passport (DPP) requirements (ECLASS) and supports Asset Administration Shell integration. **Relevance: 5/5** for the classification backbone, but licensed.

UNSPSC Segment 44 ("Office Equipment and Accessories and Supplies") Oklahoma.gov is **fully free to download** undp as an Excel file (157,116 total items, Wikipedia available in German). Wikipedia It provides a simple 8-digit hierarchical code Government of Indiana with categories like 44-10-31 (Printer and facsimile supplies, including code 44103103 for Toner) Oklahoma.gov and 44-12-17 (Writing instruments). However, UNSPSC is purely taxonomic — **no material composition properties** exist. Best used as a supplementary free classification layer. **Relevance: 3/5.**

GS1 GPC (Segment 10005000: "Stationery/Office Machinery/Occasion Supplies") is freely available Open Product Data for download in Excel, Word, and XML at gpc-browser.gs1.org. Updated twice annually with a 4-tier hierarchy (Segment → Family → Class → Brick with attributes). GS1 GS1's Verified by GS1 service allows **30 free GTIN lookups/day**; full API access costs ~€6,500/year. GS1 **Relevance: 3/5** for classification, primarily useful for linking GTIN barcodes to product categories.

Wikidata provides a completely free (CC0), SPARQL-queryable knowledge base with German labels. The critical property **P186 ("made from material")** directly supports waste disposal classification for product *types* (not individual SKUs). Example SPARQL queries can extract all subclasses of office supplies (Q15714692), writing instruments with materials, or items with GTIN barcodes. Coverage is good for generic product types but weak for specific commercial products. **Relevance: 4/5** as a foundational ontology layer.

BMEcat catalogs are closed B2B documents, but samples and specs are free

No openly downloadable, real-world BMEcat office supply catalogs exist — they are confidential B2B exchange documents. However, several resources enable working with the format.

Free BMEcat resources: The BME e.V. downloads page (bme.de) provides full specifications, XSD schemas, DTDs, and an **XML examples ZIP** (eclass_bmecat_examples.zip) for BMEcat 2005.1 — all free, no registration. The **BMECat.net GitHub repository** (github.com/stephanstapel/BMECat.net) contains GitHub 14 real-world Weidmüller industrial catalog XML files GitHub plus a demo "Office Material" catalog. GitHub The olivere/bmecat Go library repository GitHub includes test files with office supply sample data (Post-Safe Polythene Envelopes, Office Supplies AG). The BMEcat standard defines rich fields: Itscope SUPPLIER_AID, EAN/GTIN, descriptions, manufacturer info, eCl@ss/UNSPSC classification via ARTICLE_FEATURES, volume-tiered pricing, MIME_INFO for images/datasheets/safety data sheets, and catalog group hierarchies. ATAMYA

Suppliers using BMEcat: Unite/Mercateo requires BMEcat 1.2 as its primary catalog format and documents the specification publicly. Unite Simplesystem Schäfer Shop Acto GmbH offers BMEcat and Excel catalogs with eCl@ss classification (versions 4.0 through 7.1) — available to registered B2B customers only. Schäfer Shop Lyreco supports BMEcat, JSON, OCI punchout, and HTML formats for B2B integration. Lyreco Lyreco Simple System (widely used by German universities) requires suppliers to provide catalogs **exclusively in BMEcat format**. Simplesystem Bechtle confirms BMEcat, JSON, and CIF support with eCl@ss/UNSPSC classification. Bechtle

BMEcat tools: SEPIA BMEcatHub (sepia.de/en/bmecathub) imports, edits, and exports BMEcat catalogs with a free basic version. e-proCAT supports 200+ customer-specific BMEcat variations. e-proCAT nexoma CatalogExpress specifically lists "office supplies (PBS)" as a served industry.

German university procurement data is entirely closed

No German university publishes procurement catalogs as open data. Framework contracts (Rahmenverträge) are publicly tendered and award notices are published, but actual product catalogs remain confidential between contracted suppliers and purchasing departments. Documented examples include Leibniz-Gemeinschaft institutes (Lyreco, €205,342.99), Openprocurements Universität Bamberg (allpremio, €137,918.64), Openprocurements and NRW state framework (Vergabe.NRW, effective 01.01.2024, with Lyreco and Isfort).

The Kaufhaus des Bundes (KdB) at kdb.bund.de is Wikipedia accessible only to registered federal agencies — no product catalog data is publicly available. E-procurement portals (cosinex, Vergabe24, DTVP) contain tender notices with CPV codes (30192700 = Büromaterial) but no product-level data. Patterno The only public procurement data sources are tender aggregators: openprocurements.com (free, CPV-coded), Stotles (EU/UK notices), and Vergabe24 (570,000+ tenders/year, 80,000+ companies). LinkedIn

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Material composition is the critical unsolved gap

No publicly accessible database systematically catalogues the full material composition of office products (e.g., "this pen is 40% polypropylene, 30% polystyrene, 20% brass, 10% ink"). This data exists only in fragmented form:

- **ECHA SCIP:** Hazardous substances only (>0.1% w/w), no bulk download
- **Safety Data Sheets:** Only for chemical products (toner, markers, adhesives, correction fluid) — not for paper, pens, folders. Section 3 provides structured composition with CAS numbers and concentration ranges. SafetyDataSheetAPI SDS Manager offers a paid API (16M+ SDS, JSON/XML output) that auto-extracts all 16 SDS sections
- **eCl@ss properties:** Material declaration aspects added in Release 16.0, ECLASS but these are schema definitions — actual data lives in systems like PBSeasy
- **Wikidata P186:** Generic material links for product types, not commercial SKUs
- **Manufacturer declarations:** Rarely public; no aggregated source

For non-chemical products (standard paper, plastic pens, binders), **no automated source exists** — manual research from manufacturer environmental reports and sustainability declarations may be the only option. This is the single largest gap for the Fessi knowledge graph.

Integration recommendations: priority order for ETL development

Phase 1 — Immediate (free, structured, high-value):

1. **EU Ecolabel ECAT CSV** — Download from data.europa.eu; parse EAN13 codes, product names, company data, group classifications. Zero-effort acquisition, immediate knowledge graph population.
2. **Blauer Engel Fo.KUS XML** — Email blauerengel@fo-kus.info to request XML interface access. (My Sitetitle) Map DE-UZ product group standards to disposal categories.
3. **UNSPSC Segment 44 Excel** — Download free classification hierarchy. Use as initial product taxonomy.
4. **Wikidata SPARQL extraction** — Query product type hierarchies with P186 (material) and P3962 (GTIN) properties. Use as ontology backbone.
5. **GS1 GPC Excel download** — Free stationery/office classification with brick-level attributes.

Phase 2 — Medium effort (free but requires scraping/parsing):

6. **OEM Safety Data Sheet scraping** — Systematically crawl Xerox's open directory, HP's document library, and Canon's SDS collection. Parse PDF Section 3 for composition and Section 13 for disposal instructions. Map to AVV codes.
7. **ECHA SCIP portal** — Systematic queries for office product categories to extract SVHC data for markers, adhesives, toner, correction fluids.
8. **Icecat Open Catalog API** — Register for free tier; extract toner/ink cartridge datasheets with EAN, compatibility, and specifications. (Iceclog)
9. **BMEcat sample files** — Use BME.de examples and GitHub repositories to build and test BMEcat parser infrastructure.

Phase 3 — Negotiated access (highest data richness):

10. **PBSeasy (PBS Network)** — Negotiate academic/research access. This single source could provide the product catalog backbone with 1M+ eCl@ss-classified office products.
11. **Unite/Mercateo Content API** — Apply for partner program access; extract eCl@ss-classified product data with SDS attachments.
12. **eCl@ss license** — Explore university/public-institution pricing for Segment 24 classification with material declaration properties.
13. **SDS Manager API** — Evaluate paid subscription for structured extraction of toner/marker/adhesive composition data from 16M+ SDS.

Phase 4 — Specialized enrichment:

14. **FSC API** — Register at api.fsc.org (fsc) for certificate verification (Fsc) (paper supply chain).
15. **EPEAT CSV export** — Extract printer/copier/MFD environmental ratings.
16. **TRI Resources T3 API** — Evaluate paid subscription for comprehensive cartridge compatibility cross-referencing.
17. **GESTIS substance linking** — Map toner/ink component CAS numbers to GESTIS hazard profiles (deep-link integration). (DGUV)

Gaps that remain unfilled

Material composition for non-chemical products is the most critical gap. No database provides structured material breakdown for standard office products like pens, folders, binders, staplers, or scissors. This affects the core Fessi use case of "what bin does this go in?" for everyday items.

Cartridge-to-AVV-code mapping does not exist as a pre-built dataset. The multi-layer pipeline (identify → find SDS → extract composition → check thresholds → assign AVV code) must be constructed from scratch.

Packaging composition data is almost entirely absent from all sources investigated. No database systematically documents how office products are packaged (blister pack vs. cardboard vs. shrink wrap) and what the packaging is made of.

University-specific product lists are closed. Without access to a specific university's procurement system (likely Simple System or JAGGAER with BMEcat catalogs), Fessi cannot know which exact products are in use. A partnership with the university's procurement department is essential.

Disposal instructions in structured form are rare. AVV codes exist as a legal framework, and some SDS include Section 13 disposal guidance, but no source provides machine-readable "put this in the yellow bin" instructions mapped to specific product types. This is the knowledge that Fessi must ultimately generate — making it the knowledge graph's unique intellectual contribution rather than something importable from external sources.